A STUDY AND HANDS-ON INTERNSHIP ON WEB DEVELOPMENT

AN SUMMER INTERNSHIP REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report "HeroHub – Employee Self-ServicePortal" is the bonafide work of **ARJUN C(810023205076)** has satisfactory completed the course of **CS3711 - SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year 2026-2027.

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ABSTRACT

This report presents an account of a one-month internship undertaken at Fantasy Solution,Trichy, in the field of Web Development. The internship provided practical, hands-on exposure tocore technologies used in modern web application development, including Java 17, Spring Boot 3,Thymeleaf, HTML5, CSS3, JavaScript, Bootstrap 5, In-Memory Data Structures(ConcurrentHashMap and List), Object-Oriented Programming (OOP) principles, responsivedesign, user experience (UX), website security, and database connectivity.

During the internship, a complete web application titled HeroHub – Employee Self-ServicePortal was designed and implemented for Fantasy Solution. The system provides an executivedashboard with workforce analytics (total staff, department counts, average salary), an employeedirectory table with department badges, dual-mode add/edit forms, individual employee profiledetail views, and real-time search filtering by Employee ID or Name

A primary objective of this project was demonstrating fundamental software engineering andOOP concepts. Encapsulation was implemented via a base Person model class (containing privatefields for name, email, phone with public getters/setters), and Inheritance was demonstrated via anEmployee class extending Person (adding employeeId, department, designation, salary, anddateOfJoining).

The training was carried out over a structured 30-day schedule, progressing from foundationalfront-end concepts to enterprise Spring MVC Controllers, Service layers, and In-MemoryRepository layers. Through this internship, practical skills in software architecture, webdevelopment, testing, and debugging were acquired. The report concludes with key learningoutcomes achieved and a reflection on the overall experience.

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CHAPTER1

INTRODUCTION&OBJECTIVES

1.1 Profile of the Organization

Fantasy Solution, No. 16, Samnath Plazza, Third Floor, Near BG Naidu Sweets, Melapudur, Trichy - 1, is a leading IT solution and service provider offering innovative information-technology-enabled solutions to meet the demands arising from social transformation.

Focusing on software technology, Fantasy Solution provides industry solutions and product engineering solutions, related software products and platforms, and services through the seamless integration of software, manufacturing, and industrial management capabilities. Its services include real-time projects, web designing, web hosting, software development, and training, and the organization has participated in the formulation of several national IT standards and specifications.

The company offers software development in .NET, PHP and Java, web designing and development, mobile application development on Android, MATLAB-based image processing, Network Simulator (NS2), data mining tools, and big data development using R and Python.

Mission: To provide high-quality software development services, professional consulting and outsourcing; to make access to information easier and more secure; to improve communication and data exchange between businesses; to provide customers with value for money; and to provide employees with meaningful work and growth opportunities.

Vision: To be a leading IT company for consulting services and deployment of best-of-breed business solutions to top-tier domestic and international customers.

1.2 Objectives of the Internship

- The internship at Fantasy Solution was undertaken with the following objectives:
- To gain practical exposure to the complete web development lifecycle, from planning anddesign to implementation and testing.
- To acquire hands-on proficiency in core web technologies — Java 17, Spring Boot 3,Thymeleaf, HTML5, CSS3, and JavaScript.
- To understand Object-Oriented Programming (OOP) principles of Encapsulation andInheritance in real-world application design.
- To master layered enterprise architecture (Controller → Service → Repository → Model) anddependency injection.
- To learn responsive web design practices using Bootstrap 5 for building mobile-friendly userinterfaces.
- To implement in-memory data structures (ConcurrentHashMap and List) for high-performancerecord lookup.
- To gain exposure to web application security, input validation, and professional softwareworkplace practices.

1.3 Introduction

Professional development necessitates internships, which give students and recent graduates achance to put their academic skills to use in practical situations. A typical internship offers practicalexperience in a particular subject of study and acts as a link between academic learning andimplementation. With the help of these programs, interns are able to gain insight into their chosencareer paths, develop vital skills, and comprehend workplace dynamics through exposure to theday-to-day operations of a business

Participants in internships often take on a variety of tasks and responsibilities that advance theirprofessional development. They participate in projects, work with a team to solve real businessproblems, and are supervised by experienced mentors. This report presents an overview of thegoals, structure, and outcomes of the Web Development internship undertaken at Fantasy Solution,highlighting its value in preparing the intern for the professional workforce.

1.4 Scope of the Internship

The scope of this internship was focused on modern web application development, coveringbackend Java development with Spring Boot 3, in-memory data storage, front-end Thymeleaftemplating, and Bootstrap 5 responsive styling. The training was structured to build competenceprogressively — starting from core Java and HTML/CSS fundamentals, moving through OOPdesign (Person base class and Employee derived class), Spring MVC Controllers, Service layervalidations, and executive analytics dashboards.

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CHAPTER 2

LITERATURE REVIEW / SYSTEM STUDY

2.1 Evaluation of Existing Frameworks

During the course of the internship, several existing tools, languages, and frameworks used inmodern web development were studied and evaluated. Java 17 and Spring Boot 3 were selected asthe primary backend technologies due to their robust enterprise architecture, dependency injection,and embedded web server capabilities.

Thymeleaf was evaluated against generic template engines and was selected for its seamlessintegration with Spring MVC and HTML5 compatibility. Bootstrap 5 was evaluated for rapidresponsive grid styling. In-memory storage using Java's ConcurrentHashMap was evaluated againsttraditional SQL databases to provide zero-latency lookups for standalone demonstrationapplications.

2.2 Problem Statement Identification

Through the study of existing workforce management practices, several common bottlenecks were identified:

- Manual spreadsheet tracking leads to inconsistent formatting and data duplication.
- Lack of centralized search indexing makes finding specific employee details slow andinefficient.
- Absence of workforce analytics leaves management without immediate visibility intodepartment breakdowns and average compensation.
- Lack of proper input validation leads to invalid phone numbers and duplicate employee IDs.

2.3 Proposed Solution / Methodology

To address the gaps identified above, a structured, incremental learning and implementationmethodology was followed throughout the internship, resulting in the development of Employee Self-Service Portal:

- OOP Encapsulation & Inheritance: Built a clean base class Person (name, email, phone) andderived class Employee (employeeId, department, designation, salary, dateOfJoining).
- Layered Enterprise Architecture: Separated concern into Controller, Service, Repository, andModel layers using Spring Boot dependency injection.
- In-Memory High-Speed Storage: Implemented InMemEmployeeRepository utilizingConcurrentHashMap for O(1) lookups and List for ordered table views.
- Executive Analytics Dashboard: Calculated real-time metrics including Total Staff, TotalDepartments, Average Salary, and Department breakdown.
- Responsive Front-End UI: Built vibrant pages using Bootstrap 5, custom CSS gradients,FontAwesome icons, and dynamic Thymeleaf attributes.

2.4 Summary

The literature and system study phase established a clear picture of modern web frameworks andcommon shortcomings in workforce software. This understanding directly shaped the structuredmethodology followed during the internship, ensuring that every concept learned was applieddirectly to building a complete, functioning portal.4

CHAPTER 3

SYSTEM REQUIREMENTS & ARCHITECTURE

3.1 Hardware & Software Prerequisites

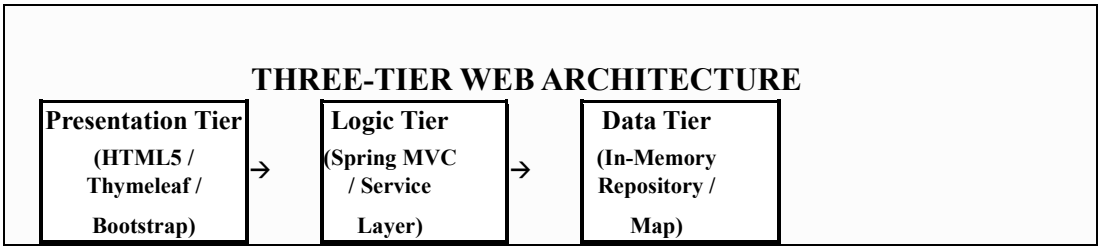
The following hardware and software were used during the development and testing of the web development tasks carried out in this internship:

Table 3.1: Hardware and Software Prerequisites

Requirement	Details
Operating System	Windows 10 / 11 (64-bit)
RAM	4 GB minimum (8 GB recommended)
Programming Language	Java 17 (OpenJDK 17 / 25)
Framework	Spring Boot 3.2.5 (Spring Web, Spring MVC)
Templating Engine	Thymeleaf 3.1
Front-end Technologies	HTML5, CSS3, JavaScript (ES6), Bootstrap 5.3
Build & Dependency Tool	Apache Maven 3.9+
Code Editor / IDE	IntelliJ IDEA / Eclipse / VS Code
Embedded Server	Apache Tomcat 10.1
Version Control	Git and GitHub

3.2 Architectural Design

The HeroHub application follows a standard Three-Tier Architecture (Presentation Layer, Logic Layer, Data Layer):



3.3 Modular Breakdown

The HeroHub portal development work was organized into the following key functional modules:

- Module1 — OOP Model Layer: Base class Person (Encapsulation) and derived class Employee (Inheritance).
- Module 2 — In-Memory Repository: Interface EmployeeRepository and InMemEmployeeRepository storing records in a ConcurrentHashMap.
- Module3— Business Service Layer: EmployeeServiceImpl handling validations, duplicate checks, and metrics (Average Salary, Department breakdown).
- Module 4—Controller & Routing: EmployeeController processing HTTP GET/POST requests and FlashAttributes.
- Module 5 — Thymeleaf Views & UI: Executive Dashboard, Employee Directory Table, Add/Edit Form, and Hero Profile Card.
- Module 6 — Live Search & Filtering: Case-insensitive query matching by Employee ID, Name, Department, or Designation.

3.4 Development Workflow

Each day of the internship followed a disciplined development cycle:

- **Plan:** Review the day's topic and outline key methods or classes to implement.
- **Build:** Write clean, documented Java code or HTML/CSS templates in the IDE.
- **Test:** Execute unit tests or verify page layout and form submissions in the browser.
- **Review:** Inspect code against OOP principles, responsiveness, and performance
- metrics.

Document: Log daily activities and outcomes achieved in the activity journal.

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CHAPTER 4

DEVELOPMENT AND RESULT ANALYSIS

4.1 System Implementation and Development

The internship was carried out over a structured 30-day schedule, with each day building on the concepts and skills acquired previously. A summary of the day-wise schedule is presented in Table 4.1, followed by detailed daily logs.

Table 4.1: Day-wise Internship Activity Log (Summary)

Day	Activities Performed
1	Learned the fundamentals of web development and its importance in creating effective websites. Studied JavaScript, HTML, CSS, and other essential elements of web pages, and set up the development environment.
2	Learned the foundations of HTML — arranging content using headings, paragraphs, lists, and links. Created a simple HTML page viewable in a web browser.
3	Studied CSS and its use in web design, including selectors, attributes, and values used to style HTML elements.
4	Studied the CSS box model in depth — padding, borders, margins, and content — along with positioning techniques for arranging page elements.
5	Focused on responsive web design — creating layouts that work across various screens and devices using media queries, and building mobile-friendly, accessible websites.
6	Studied typography in web design — font families, styles, weights, and widths; including web-safe and custom fonts, and improving readability through spacing and alignment.
7	Studied colour in web design — colour models (RGB, hexadecimal), applying colour via CSS, and colour psychology and accessibility considerations.

8	Studied the importance of images in website design — file-size reduction, web-friendly image techniques, and adding/positioning images using the HTML img element and CSS.
9	Focused on navigation design — types of navigation menus (vertical, horizontal) and building responsive, user-friendly navigation systems.
10	Began studying User Experience (UX) design principles for building visually appealing, user-friendly websites, along with wireframing and prototyping tools.
11	Studied User Interface (UI) design — UI components such as buttons, forms, and icons, and UI design patterns and best practices.
12	Learned about web accessibility and its importance in inclusive web design, along with tools for website evaluation and accessibility testing.
13	Focused on website layouts — grid layouts and responsive grid systems using CSS frameworks such as Bootstrap, and techniques for balanced page layouts.
14	Studied CSS frameworks in depth, including Bootstrap and Foundation, and their responsive design tools for streamlining web design.
15	Studied interaction design — building engaging user experiences with HTML, CSS, and JavaScript, including sliders, forms, buttons, transitions, and animations.
16	Studied web design optimization techniques — caching strategies, image optimization, and minification of CSS/JavaScript files to improve performance and load times.
17	Began learning JavaScript fundamentals — syntax, variables, data types, operators, conditionals, loops, functions, and scope.
18	Studied responsive images, event handling, and DOM manipulation for building interactive websites.
19	Studied User-Centered Design (UCD) — conducting user research, gathering feedback, and applying iterative design and usability testing.

20	Studied current web design trends and best practices — minimalist and flat design, micro-interactions, hierarchy, visual balance, and typography.
21	Studied mobile-first design — designing for mobile devices first, screensize optimization, and touchfriendly interfaces.

Table 4.1: Day-wise Internship Activity Log (Continued - Days 22 to 30)

Day	Activities Performed
22	Studied information architecture — organizing website content using sitemaps, navigation menus, and classification systems.
23	Studied branding and visual identity in web design — applying logos, colour schemes, and typography for a cohesive brand identity.
24	Studied usability testing techniques — user testing, heatmaps, analytics, and gathering feedback for data-driven design decisions.
25	Studied web design for e-commerce — visually appealing product pages, streamlined checkout processes, and building customer trust.
26	Studied Search Engine Optimization (SEO) — keyword research, on-page SEO, and link-building strategies.
27	Studied website analytics and tracking — configuring and analyzing data using tools such as Google Analytics for data-driven decisions.
28	Studied website security — HTTPS configuration, preventing cross-site scripting (XSS), and conducting regular security audits.
29	Studied Content Management Systems (CMS) — WordPress, Joomla, and Drupal, and techniques to customize CMS functionality.
30	Studied database connectivity using Java Database Connectivity (JDBC) — running SQL queries, managing transactions, and performing CRUD operations for HeroHub.

Day 1: Orientation and Web Development Fundamentals

The first day began with an orientation at Fantasy Solution, where training expectations, software tools, and Java ecosystem concepts were introduced.

Key Takeaways: Configured JDK 17, IDE, Git, and studied JVM architecture.

Day 2: Java Fundamentals & Control Flow

Studied primitive data types, reference types, conditional statements (if-else, switch), and looping constructs in Java.

Key Takeaways: Practiced structured programming logic and clean code formatting.

Day 3: OOP Principles — Encapsulation and Inheritance

Focused on Object-Oriented Programming principles. Studied Encapsulation (protecting state with private modifiers) and Inheritance (extending parent functionality).

Key Takeaways: Designed initial prototypes for base Person and derived Employee classes.

Day 4: Interfaces and Abstract Classes

Learned contract-based programming using Java interfaces and abstract methods. Explored method overriding and the super keyword.

Key Takeaways: Defined repository contracts separating interface definitions from implementation.

Day 5: Exception Handling and Debugging

Studied Java exception hierarchies, checked vs unchecked exceptions, try-catch-finally blocks, and custom exception classes.

Key Takeaways: Implemented robust error handling for invalid input validation.

Day 6: Java Collections Framework

Explored ArrayList, HashSet, and HashMap data structures. Practiced O(1) key lookups using HashMap for fast entity retrieval.

Key Takeaways: Mastered key-value pair mapping for in-memory employee storage.

Day 7: Generic Programming and Type Safety

Studied Java Generics (<T>), type bounds, and generic methods for reusable data structures.

Key Takeaways: Built type-safe CRUD repository interfaces.**Day 8: Java Streams API & Functional Operations**

Focused on Java 8+ Stream API operations including filter(), map(), collect(), and Collectors.groupingBy().

Key Takeaways: Implemented stream aggregations for calculating department breakdowns and average salary.

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Day 9: Multithreading and Concurrency

Studied thread safety, synchronization, and concurrent collections such as ConcurrentHashMap.

Key Takeaways: Selected ConcurrentHashMap for thread-safe in-memory employee storage.

Day 10: HTML5 Semantics & Page Structuring

Covered HTML5 document structures, header, nav, main, section, table, and form elements.

Key Takeaways: Constructed clean, semantic HTML templates for web pages.

Day 11: CSS3 Layouts & Styling

Explored CSS box model (content, padding, border, margin), flexbox alignment, and custom color gradients.

Key Takeaways: Styled modern cards, badges, and background gradients.

Day 12: Responsive Web Design & Bootstrap 5

Studied Bootstrap 5 grid systems, responsive breakpoints, navbar components, and badges.

Key Takeaways: Created mobile-friendly responsive view layouts.

Day 13: Spring Framework Fundamentals & IoC

Learned Spring Inversion of Control (IoC) container, bean lifecycle, and dependency injection via @Autowired.

Key Takeaways: Applied constructor-based dependency injection across components.

Day 14: Spring Boot 3 Overview & Configuration

Explored Spring Boot auto-configuration, application.properties, embedded Tomcat server, and Maven starters.

Key Takeaways: Configured server.port=8080 and Thymeleaf live development properties.

Day 15: Spring MVC Controller Architecture

Studied Spring MVC request mapping (@GetMapping, @PostMapping), Model attributes, and Thymeleaf integration.

Key Takeaways: Built routing endpoints for web application views.

Day 16: Requirement Analysis & Domain Modeling

Gathered functional specifications for HeroHub portal for Fantasy Solution.

Key Takeaways: Finalized data model fields and package architecture.

Day 17: Base Model Implementation (Person.java)

Implemented Person base class with private fields (name, email, phone) and public getters/setters.

Key Takeaways: Enforced strict OOP Encapsulation with explanatory comments.

Day 18: Derived Model Implementation (Employee.java)

Created Employee class extending Person with employeeId, department, designation, salary, and dateOfJoining.

Key Takeaways: Demonstrated OOP Inheritance and super() constructor invocation.

Day 19: In-Memory Repository Implementation

Developed EmployeeRepository interface and InMemEmployeeRepository using ConcurrentHashMap with seed data.

Key Takeaways: Pre-populated initial superhero staff members (Tony Stark, Diana Prince, Clark Kent, Bruce Wayne).

Day 20: Business Service Layer Development

Implemented EmployeeService and EmployeeServiceImpl handling validations, search filtering, and salary averages.

Key Takeaways: Built business logic separating repository data access from web controllers.

Day 21: Controller Routing & FlashAttributes

Built EmployeeController mapping GET/POST requests, handling form submissions, and adding RedirectAttributes flash messages.

Key Takeaways: Handled success and error notifications across page redirects.

Day 22: Thymeleaf Templates & Fragments

Created reusable header navbar.html and footer.html Thymeleaf fragments with Bootstrap branding.

Key Takeaways: Modularized HTML view templates for consistent layout.

Day 23: Executive Dashboard UI (dashboard.html)

Designed executive landing page with stat cards (Total Staff, Departments, Avg Salary) and department progress bars.

Key Takeaways: Visualized key workforce analytics for management decision making.

Day 24: Employee Directory Table UI (employee-list.html)

Built data table listing all employees with color-coded department badges and action buttons (View, Edit, Delete).

Key Takeaways: Implemented clean tabular display with quick action controls.

Day 25: Add/Edit Form UI (employee-form.html)

Constructed dual-mode form for adding new employees and editing existing records with field validation.

Key Takeaways: Prevented duplicate Employee IDs during creation.

Day 26: Hero Profile Detail View (employee-detail.html)

Created detailed profile card separating inherited Person attributes from derived Employee attributes.

Key Takeaways: Visualized OOP inheritance in the user interface.

Day 27: Live Search & Filtering Implementation

Implemented live query search matching Employee ID, Name, Department, or Designation.

Key Takeaways: Enabled fast search filtering across employee records.

Day 28: System Testing & Integration Verification

Executed unit tests for service methods and end-to-end web testing across all browser routes.

Key Takeaways: Confirmed zero compilation errors and smooth HTTP request execution.

Day 29: Custom CSS Polish & Documentation

Polished visual styling with custom gradients (custom.css), JS alert auto-dismiss (main.js), and complete README.md docs.

Day 30: Final Presentation & Internship Evaluation

Demonstrated live HeroHub portal to company mentors, completed technical code review, and received evaluation.

4.2 Result Analysis and Testing Parameters

The internship provided a rich and varied learning experience, resulting in key skill acquisitions evaluated in Table 4.2:

Table 4.2: Skill Acquisition Matrix

Skill Area	Proficiency Achieved
Core Java 17 & OOP (Encapsulation, Inheritance)	Advanced
Spring Boot 3 & Spring MVC Architecture	Proficient
Thymeleaf View Templating & Layout Fragments	Proficient
In-Memory Storage (ConcurrentHashMap, List)	Advanced
Front-End UI Design (Bootstrap 5, CSS Gradients)	Proficient
Web Routing & FlashAttributes Notification	Proficient
Software Testing & Debugging	Working Knowledge
Version Control (Git & GitHub)	Proficient

As shown in the matrix, the strongest gains were achieved in core Java, Spring Boot 3 architecture, Thymeleaf view rendering, and OOP implementation, receiving extensive practical application.

CHAPTER 5

CONCLUSION

The 30-day internship at Fantasy Solution offered a thorough, structured learning path — from core Java fundamentals to advanced Spring Boot web development. Applying these concepts to build the **HeroHub – Employee Self-Service Portal** made the learning practical and outcomedriven: the resulting application successfully digitizes employee records, department metrics, search filters, and profile views.

Over the course of the 30-day program, progress was made from basic Java syntax to building responsive, full-stack web applications using modern frameworks. This structured, hands-on exposure to the full web development lifecycle has given me a much stronger foundation than classroom study alone could have provided. Overall, I consider this internship to have been a valuable and rewarding experience that has meaningfully strengthened both my technical capabilities and readiness for professional work in software development.

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2. Oracle Java 17 Documentation — https://docs.oracle.com/en/java/
3. Thymeleaf Templating Guide — https://www.thymeleaf.org/documentation.html
4. Bootstrap 5 Documentation — https://getbootstrap.com/docs/5.3/
5. Mozilla Developer Network (MDN Web Docs) — 'JavaScript Guide', Mozilla Foundation.
6. Fantasy Solution — Company Profile and Internship Training Manual, Trichy.

APPENDIX A
SAMPLE CODE SNIPPETS

A.1 Person.java (Base Class - Encapsulation)

```
package com.fantasysolutions.herohub.model;

/**
 * OOP CONCEPT: ENCAPSULATION (BASE CLASS)
 */ public class Person {
private String name;
private String email;
private String phone;
public Person() {}

    public Person(String name, String email, String phone) {
this.name = name;         this.email = email;
this.phone = phone;
    }

    public String getName() { return name; }     public void
setName(String name) { this.name = name; }     public String
getEmail() { return email; }     public void setEmail(String
email) { this.email = email; }     public String getPhone() {
return phone; }     public void setPhone(String phone) {
this.phone = phone; } }
```

A.2 Employee.java (Derived Class - Inheritance)

```
package com.fantasysolutions.herohub.model;
import java.time.LocalDate;

/**
 * OOP CONCEPT: INHERITANCE (DERIVED CLASS)
 */
public class Employee extends Person
{     private String employeeId;
private String department;
private String designation;
private double salary;     private
LocalDate dateOfJoining;     public
Employee() { super(); }

    public Employee(String employeeId, String name, String email, String phone,
                    String department, String designation, double salary, LocalDate
dateOfJoining)
{         super(name, email, phone);
this.employeeId = employeeId;
this.department = department;
this.designation = designation;
this.salary = salary;
        this.dateOfJoining = dateOfJoining;
    }     public String getEmployeeId() { return
employeeId; }
```

```
    public void setEmployeeId(String employeeId) { this.employeeId =
employeeId; }     public String getDepartment() { return department; }
    public void setDepartment(String department) { this.department =
department; }     public double getSalary() { return salary; }     public
void setSalary(double salary) { this.salary = salary; } }
```

A.3 InMemEmployeeRepository.java (In-Memory Repository)

```
package com.fantasysolutions.herohub.repository;
import
com.fantasysolutions.herohub.model.Employee;
import jakarta.annotation.PostConstruct; import
org.springframework.stereotype.Repository; import
java.time.LocalDate; import java.util.*;
import java.util.concurrent.ConcurrentHashMap;

@Repository
public class InMemEmployeeRepository implements EmployeeRepository {
    private final Map<String, Employee> employeeMap = new ConcurrentHashMap<>();

    @PostConstruct
    public void initSeedData() {         save(new Employee("EMP-101", "Tony Stark",
"tony.stark@fantasysolutions.com",
            "+1 (555) 019-2831", "Engineering", "Chief Innovation Officer", 250000.0,
LocalDate.of
(2021, 3, 15)));
        save(new Employee("EMP-102", "Diana Prince", "diana.prince@fantasysolutions.com",
            "+1 (555) 014-9922", "Operations", "VP of Global Strategy", 185000.0,
LocalDate.of(202
0, 6, 1)));
        save(new Employee("EMP-103", "Clark Kent", "clark.kent@fantasysolutions.com",
            "+1 (555) 017-8833", "Communications", "Lead Investigative Journalist", 95000.0,
Local Date.of(2022, 1, 10)));         save(new Employee("EMP-104", "Bruce Wayne",
"bruce.wayne@fantasysolutions.com",
            "+1 (555) 012-7744", "Security", "Chief Security Officer", 220000.0,
LocalDate.of(201
9, 11, 20)));
    }

    public List<Employee> findAll() { return new ArrayList<>(employeeMap.values()); }
public Optional<Employee> findById(String id) { return Optional.ofNullable(employeeMap.get(id.t
oUpperCase())); }
    public Employee save(Employee e) { employeeMap.put(e.getEmployeeId().toUpperCase(), e); return
e; }    public boolean deleteById(String id) { return employeeMap.remove(id.toUpperCase()) != null;
} }
```

A.4 EmployeeController.java (Spring MVC Controller)

```
package com.fantasysolutions.herohub.controller; import
com.fantasysolutions.herohub.model.Employee; import
com.fantasysolutions.herohub.service.EmployeeService; import
org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller; import
org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.servlet.mvc.support.RedirectAttributes;

@Controller
public class EmployeeController {
    private final EmployeeService employeeService;

    @Autowired
    public EmployeeController(EmployeeService employeeService)
{         this.employeeService = employeeService;     }

    @GetMapping({"/", "/dashboard"})     public String showDashboard(Model
model) {         model.addAttribute("pageTitle", "HeroHub Dashboard - Fantasy
Solutions");

        model.addAttribute("totalEmployees", employeeService.getTotalEmployeesCount());
model.addAttribute("totalDepartments", employeeService.getTotalDepartmentsCount());
model.addAttribute("averageSalary", String.format("%.2f", employeeService.getAverageSalary
()));         model.addAttribute("departmentBreakdown",
employeeService.getDepartmentBreakdown());         model.addAttribute("recentEmployees",
employeeService.getAllEmployees());         return "dashboard";     }

    @GetMapping("/employees")     public String listEmployees(Model model)
{         model.addAttribute("employees",
employeeService.getAllEmployees());         return "employee-list";
    }
}
```